The DICTO Metaxyhexecontahedron

The solid in the last gap of the golden cycle: a dodecahedron–icosahedron hull whose icosahedron edge is $9/\varphi^4$ times its dodecahedron edge

DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com
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Licence CC BY 4.0 · An introduction to one solid; its parent paper: Vajrayana, DICTO DOI: 10.5281/zenodo.23429911. Data and checks are included with this record.
The idea that one more solid fits the last gap of the cycle, and the name, are DICTO's; the exact computations and the text were done with an AI assistant (Claude, Anthropic).

Abstract. The DICTO Metaxyhexecontahedron M (Greek *metaxý*, "in-between", and 60 faces) is the convex hull of a regular icosahedron and a dodecahedron placed over its faces, at the one size where the icosahedron edge is exactly $9/\varphi^4 \approx 1.31308$ times the dodecahedron edge. In closed form it is the hull of $(0, \pm1, \pm\varphi)$ and $(\varphi^5/9)\cdot\{(\pm1, \pm1, \pm1), (0, \pm\varphi, \pm1/\varphi)\}$ (cyclic). It has 32 corners, 90 edges and 60 congruent isosceles triangles, an insphere but no midsphere. In the twelve-station golden cycle of Vajrayana it is the station between the rhombic triacontahedron and the next icosahedron: its 12 five-fold corners are the triacontahedron's own, its 20 three-fold corners sit exactly on the centres of the next icosahedron's faces, and each of the two layers it makes is a single kind of piece. Everything here is exact in $\mathbb{Q}(\sqrt{5})$.

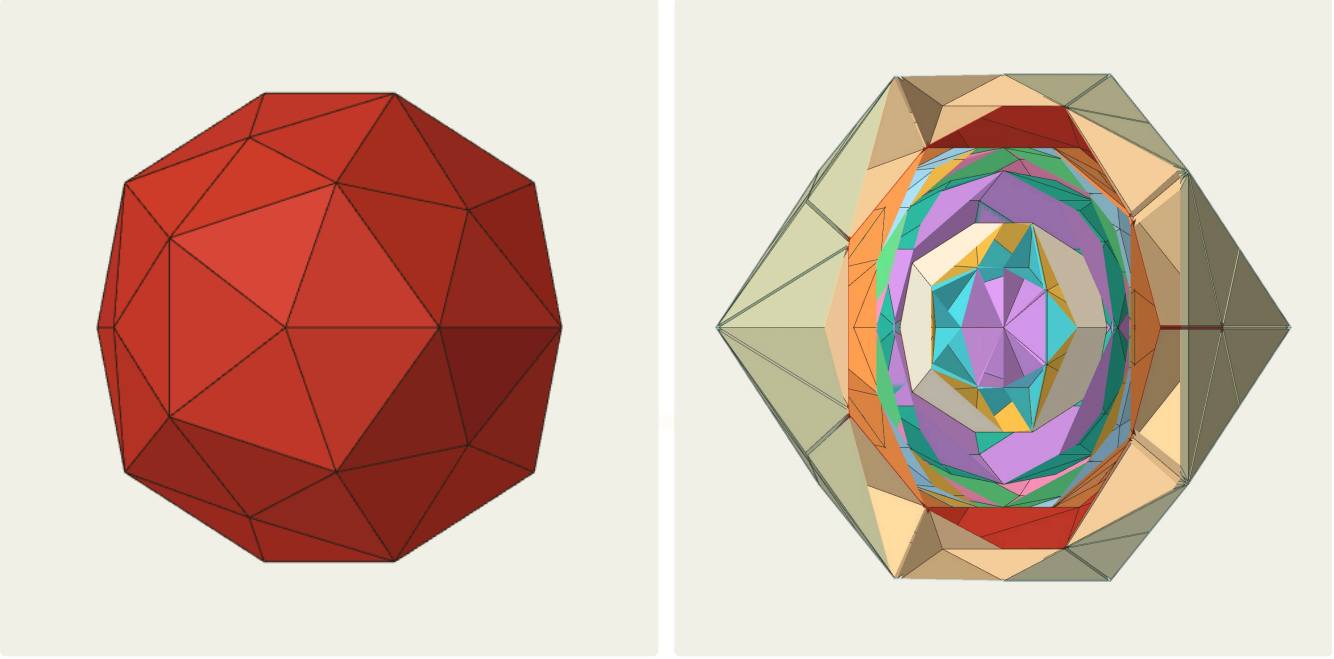


Figure 1. Left: the DICTO Metaxyhexecontahedron. Right: in the twelve-station cycle (brick red), between the rhombic triacontahedron (orange) and the next icosahedron (glass), with the 60 pieces of one kind that fill the step outside it (sand). Drawn from exact corners.

1. The solid

Take the icosahedron with corners $(0, \pm1, \pm\varphi)$ and their cyclic shifts (edge 2), and the dodecahedron with corners $(\pm1, \pm1, \pm1)$ and the cyclic shifts of $(0, \pm\varphi, \pm1/\varphi)$, placed so that its corners lie over the icosahedron's face centres (edge $2/\varphi$). Scale the dodecahedron by k

and take the convex hull. Its icosahedron edge is then φ/k times its dodecahedron edge. The DICTO Metaxyhexecontahedron is the member with $k = \varphi^5/9 = 11/18 + 5\sqrt{5}/18$, so that the ratio is $9/\varphi^4$ (squared, $81/\varphi^8$).

Property	Value (exact)
corners	32: 12 five-fold (the icosahedron's), 20 three-fold (the dodecahedron's)
faces, edges	60 congruent isosceles triangles, 90 edges; the combinatorics of the pentakis dodecahedron
squared edge lengths	$94/81 + 14\sqrt{5}/27$ (60 edges, five-fold to three-fold) and $14/9 + 2\sqrt{5}/27$ (30 edges, three-fold to three-fold)
squared corner radii	$5/2 + \sqrt{5}/2$ (five-fold) and $41/18 + 55\sqrt{5}/54$ (three-fold)
insphere	all 60 faces at squared distance $7249/4162 + 16209\sqrt{5}/20810$
midsphere	none: the two kinds of edge lie at squared distances $2 + 20\sqrt{5}/27$ and $161/81 + 8\sqrt{5}/9$
volume	$410/27 + 550\sqrt{5}/81 \approx 30.3684$ (in these units)

Table 1. The DICTO Metaxyhexecontahedron in closed form (`scripts/hull_family_exact.py`, every face plane and every corner tested exactly).

2. Where it comes from: the last gap of the golden cycle

The Spheres of Dharma (DICTO DOI: 10.5281/zenodo.23310833) nest icosahedral shells in a cycle that repeats every φ^3; Progenitors (DICTO DOI: 10.5281/zenodo.23413184) set the DICTO Ogdoecontahedron into it, and Vajrayana (DICTO DOI: 10.5281/zenodo.23429911) closed its loose steps. In the step from the rhombic triacontahedron RT to the next icosahedron $I\cdot\varphi^3$, the triacontahedron reaches only 0.812 of the way. DICTO judged that one more solid should fit there. Among the convex icosahedral solids none touches both neighbours (Vajrayana, Table 3, float); this member of the dodecahedron–icosahedron hull family does. In the cycle's own units it is the hull of the triacontahedron's 12 five-fold corners and the 20 face centres of $I\cdot\varphi^3$ (squared radii $45/8 - 9\sqrt{5}/8$ and $47/24 + 7\sqrt{5}/8$; the same solid as Table 1, checked exactly). Cut by cones from the centre over the faces of the inner solid, the step RT → M is 30 pieces of one kind and M → $I\cdot\varphi^3$ is 60 pieces of one kind (a triangular-bipyramid shape); M takes 0.356 of the step's volume. Its polar, with 12 regular pentagons and 20 hexagons whose sides alternate in the ratio $(14 + 9\sqrt{5})/11$, closes the matching step of the polar-reversed cycle.

3. Its family

The hulls of a dodecahedron and an icosahedron in dual position form a one-parameter family with the pentakis dodecahedron's combinatorics, fully symmetric for edge ratios from $(5 + 3\sqrt{5})/10$ to φ. Its named members are the Catalan pentakis dodecahedron at $3(7 + 5\sqrt{5})/38 \approx 1.435$, the DICTO Palintriacontahedron at $3/\sqrt{5} \approx 1.342$ (DICTO DOI: 10.5281/zenodo.23430866), and the DICTO Metaxyhexecontahedron at $9/\varphi^4 \approx 1.313$. We note without explanation that $9/\varphi^4$ is also the exact gain per round of the tightest progenitor cycle in Progenitors.

4. Status

Known: the family (the pentakis dodecahedron as the hull of a dodecahedron and an icosahedron, and its fully symmetric range: Polytope Wiki; the Catalan member: MathWorld,

McCooey). **Not found, in the sources checked** (listed in Vajrayana §5 and its study's SEARCH-LOG.md): this member as a named or distinguished solid, or the $9/\varphi^4$ ratio as distinguished. The name returns no results. Every "not found" is limited to those sources.

Sources checked on the web (17 October 2026):

- **Polytope Wiki**, "Pentakis dodecahedron": the hull of a dodecahedron and an icosahedron is fully symmetric for edge ratios from $(5 + 3\sqrt{5})/10$ to φ; the only member given is $3(7 + 5\sqrt{5})/38$; neither $9/\varphi^4$ nor $3/\sqrt{5}$ appears.
- **Wikipedia**, "Pentakis dodecahedron": the canonical form only (icosahedron scaled by $(3\varphi + 12)/19$); no other member.
- **MathWorld**, "Pentakis Dodecahedron" and "Dodecahedron–Small Triambic Icosahedron Compound" (whose hull has the pentakis connectivity: a different member).
- **McCooey, Visual Polyhedra**: Pentakis Dodecahedron, Great Pentakis Dodecahedron, Archimedean–Catalan Hulls, Biscribed Solids; no such member.
- **OEIS**: the digits of $9/\varphi^4$ match no sequence.
- **Bridges 2017** and **Wassell and Benito 2012** (edge ratios of dual pairs, $\varphi/3$ and $\varphi^2/\sqrt{5}$); **arXiv** (nested golden shells, hulls of dual pairs; closest: arXiv:2008.00862, a tiling).
- **Web search** for the name: no results.

Polyhedron software libraries (17 October 2026):

- **SageMath** polytope library (`polytopes.*`): its icosahedral constructors are the icosahedron, dodecahedron, icosidodecahedron, truncated icosahedron and the other Archimedean solids, and the pentakis dodecahedron; none of the solids here.
- **netlib polyhedra database** (142 entries: Platonic, Archimedean and their duals, prisms, the 92 Johnson solids and five others): none of the solids here (the pentakis dodecahedron, file 41, is the canonical one).
- **Antiprism** 0.29 (A. Rossiter): used to recompute and check the structures independently (for Vajrayana and Progenitors, and the companion's type t3jD); its Conway-operator sweep is a generator, not a catalogue.
- **Not run:** Wolfram `PolyhedronData` (needs Mathematica) and Stella / Great Stella (R. Webb; its library beyond the uniform, Johnson and Stewart-toroid models is not listed online). No claim is made about them.

The reference library: see the Appendix.

5. Files

`scripts/hull_family_exact.py` and its output `data-hull-family-exact.txt` (this paper, exact). The cycle, the pieces and the dual: Vajrayana's record and the study at https://github.com/DICTOR-Master/kaleidohedra (branch research: studies/2026-10-16-progenitors: `gap_member.py`, `gap_pieces_exact.py`, `metaxy_dual.py`). See it in the app: Kaleidohedra, 3D+ → Vajrayana.

Appendix. The reference library searched

DICTO's reference library (private copies, with public citations in the Kaleidohedra repository's docs/SOURCES.md) holds 234 entries. The full text of the 64 entries below was searched on 17 October 2026 for the solid's construction, its key numbers and its name (pentakis / kis-dodecahedron / hexecontahedron / hull of a dodecahedron and an icosahedron;

enneacontahedron / rectified truncated icosahedron / compounds of the truncated icosahedron; chamfered or truncated triacontahedra; the four DICTO names). None contains any of the four solids introduced in these papers; the only hits are mentions of the classical pentakis dodecahedron (Kepler 1619; Messer 1995; de Graaf, van Roij and Dijkstra 2011) and of the chamfer operation (Goldberg 1937; the Goldberg-polyhedron and Fullerene articles). Not text-searched: 12 image-only scans and 158 link-only entries (copyrighted books and articles held as citations); no claim is made about them.

Author(s)	Year	Title
Luca Pacioli, Leonardo da Vinci (ill.)	1509	Divina proportione
Albrecht Dürer	1525	Underweysung der Messung mit dem Zirckel und Richtscheyt
Wenzel Jamnitzer, Jost Amman (engr.)	1568	Perspectiva corporum regularium
Johannes Kepler	1619	Harmonices mundi libri V
M. Goldberg	1937	A class of multi-symmetric polyhedra
Haresh Lalvani	1990	Transformational part-count in layered octahedral-tetrahedral truss configurations (NASA TM-102635)
P. W. Messer	1995	Stellations of the rhombic triacontahedron and beyond
M. Baake, F. Gähler	1998	Symmetry structure of the Elser-Sloane quasicrystal
T. C. Hales	1998	An overview of the Kepler conjecture
M. Baake	1999	A guide to mathematical quasicrystals
O. Ogievetsky, Z. Papadopolos	1999	On Quasiperiodic Space Tilings, Inflation and Dehn Invariants
U. Betke, M. Henk	1999	Densest lattice packings of 3-polytopes
Z. Papadopolos, O. Ogievetsky	1999	On Inflation Rules for Mosseri-Sadoc Tilings
Z. Papadopolos, O. Ogievetsky	1999	On quasiperiodic space tilings, inflation and Dehn invariants
R. V. Moody	2000	Model sets: a survey
M. Senechal	2004	Donald and the golden rhombohedra
Wikipedia contributors (2007 mirror)	2007	Kyushindo (2007 version)
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Grand antiprism and quaternions
M. Koca, M. Al-Ajmi, N. O. Koca	2009	Quaternionic representation of snub 24-cell and its dual polytope derived from E8 root system
S. Torquato, Y. Jiao	2009	Dense packings of polyhedra: Platonic and Archimedean solids
P. Kramer	2010	Gateways towards quasicrystals
A. M. Cutler, E. Schulte	2011	Regular polyhedra of index two, I
D. Frettlöh	2011	Icosahedral tilings in R3: the ABCK tilings (draft chapter)
J. de Graaf, R. van Roij, M. Dijkstra	2011	Dense regular packings of irregular non-convex particles
M. Koca, N. O. Koca, M. Al-Ajmi	2011	Branching of the W(H4) polytopes and their dual polytopes under the Coxeter groups W(A4) and W(H3) represented
M. Koca, N. O. Koca, M. Al-Barwani	2011	Snub 24-cell derived from the Coxeter-Weyl group W(D4)
T. Etzion	2011	Product constructions for perfect Lee codes
A. Garber, A. Gavrilyuk, A. Magazinov	2012	The Voronoi conjecture for parallelohedra with simply connected delta-surface
J. C. Lagarias, C. Zong	2012	Mysteries in packing regular tetrahedra
M. Baake, U. Grimm	2012	Mathematical diffraction of aperiodic structures
M. Koca, N. O. Koca, R. Koc	2012	Affine Coxeter group Wa(A4), quaternions, and decagonal quasicrystals
P. F. Damasceno, M. Engel, S. C. Glotzer	2012	Crystalline assemblies and densest packings of a family of truncated tetrahedra and the role of directional en
M. Kléman	2013	Defects in quasicrystals, revisited: I-flips, approximants, phason defects
N. Fujita, H. Takano, A. Yamamoto, A.-P. Tsai	2013	Cluster packing geometry for Al-based F-type icosahedral alloys
P.-P. Dechant, J. Wardman, T. Keef, R. Twarock	2014	Viruses and fullerenes: symmetry as a common thread?
A. Campello, G. C. Jorge, J. E. Strapasson, S. I. R. Costa	2015	Perfect codes in the lp metric

K. Deguchi, M. Nakayama, S. Matsukawa, K. Imura et al.	2015	Crystal structure of superconducting 1/1 cubic Au-Ge-Yb approximant with Tsai-type cluster
M. Baake, D. Damanik, U. Grimm	2015	What is aperiodic order?
M. Firsching	2015	Computing maximal copies of polytopes contained in a polytope
N. O. Koca, M. Koca, A. Al-Mukhaini, A. Al-Qanobi	2015	Quaternionic representations of the pyritohedral group, related polyhedra and lattices
R. Twarock, M. Valiunas, E. Zappa	2015	Orbits of crystallographic embedding of non-crystallographic groups and applications to virology
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C. Hann, J. E. S. Socolar, P. J. Steinhardt	2016	Local growth of icosahedral quasicrystalline tilings
R. M. K. Dietl, J.-H. Eschenburg	2016	The icosahedral quasiperiodic tiling and its self-similarity
A. Stergiou	2018	Bootstrapping hypercubic and hypertetrahedral theories in three dimensions
A. Garber	2019	Voronoi conjecture for five-dimensional parallelohedra
D. Poland, S. Rychkov, A. Vichi	2019	The conformal bootstrap: theory, numerical techniques, and applications
R. Twarock, A. Luque	2019	Structural puzzles in virology solved with an overarching icosahedral design principle
T. Ishimasa	2019	Icosahedral quasicrystal, 1/1 and 2/1 approximants in Zn-based ternary alloys containing Au and Yb/Tb
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral Polyhedra from D6 lattice and Danzer's ABCK tiling
A. Al-Siyabi, N. O. Koca, M. Koca	2020	Icosahedral polyhedra from D6 lattice and Danzer's ABCK tiling
M. Koca, R. Koc, N. O. Koca, A. Al-Siyabi	2020	Icosahedral tiling with dodecahedral structures
N. O. Koca, R. Koc, M. Koca, A. Al-Siyabi	2020	Dodecahedral Structures with Mosseri-Sadoc Tiles
B.-X. Lao, S. Rychkov	2023	3D Ising CFT and exact diagonalization on icosahedron: the power of conformal perturbation theory
D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss	2023	An aperiodic monotile
I. J. Dejter, L. R. Fuentes, C. A. Araujo	2025	Perfect truncated-metric codes in lattice graphs and superlattice graphs
S. Rychkov	2025	Conformal bootstrap: from Polyakov to our times
M. Kovačević	2026	An isodiametric theorem and lattice diameter-perfect codes in A3
R. Al Raisi, N. O. Koca, M. Koca, R. Koc	2026	Modified Mosseri-Sadoc tiles from D6
Wikipedia contributors	2026	Kenshiro Abbe
Wikipedia contributors	2026	Mitsusuke Harada
Wikipedia contributors	2026	n-flake (Sierpinski icosahedron)
Wikipedia contributors	2026	Fullerene
Wikipedia contributors	2026	Goldberg polyhedron